Persona-Conditioned Multi-Draft Merging: Method and Experiment Design

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Abstract

We evaluate whether conditioning each of several drafts on a *different survey subgroup*, then losslessly merging them, improves the viewpoint coverage of a single answer. The method is compared against a plain answer, against an instruction-varied merge over the same retrieved context, and against three controls that remove one ingredient each: the graph's subgroup selection, the divergence ranking, and all graph content. We state the design before the controls have reported, and we state in advance what each outcome licenses.

1 Motivation

A single LLM answer to a contested question tends to articulate the majority position and mention, at most, a gesture toward alternatives. Benchmarks in the Overton tradition score an answer by how many of a population's distinct viewpoint clusters it covers well enough that members of that cluster would rate it highly.

Our prior method (*merge_v2*) retrieves divergent subgroup pairs from a viewpoint knowledge graph, writes several drafts that all see the same retrieved context under different instructions, and merges them. Under a judge calibrated against human ratings it is *worse* than a plain answer on four of five archived runs (-0.075 to -0.108 coverage, $p \leq .016$), and no component of it—hierarchy, curvature, divergence ranking, anchor spread, coverage-greedy selection—is distinguishable from its own ablation.

The decomposition explains why. Writing coverage as a ledger against the plain answer, merge_v2 *gains* 0.37 clusters per question and *loses* 0.91, and what it loses is more prevalent than what it gains. It trades majority coverage for minority coverage, and the benchmark weights all clusters equally, so the trade nets negative by construction.

One arm does not behave this way. Conditioning each draft on a different subgroup—*persona_merge*—gains 0.72 and loses 0.36, with gains and losses at the same prevalence (0.499 vs. 0.504). It adds without shedding. It beats the plain answer under both judges we have used ($+0.075$, $p = .002$; $+0.060$, $p = .024$), making it the only arm whose sign survives the judge swap, and it does so while producing answers $6.6\times$ longer than the plain answer under a judge that penalises length.

That result rests on one generation run and on an arm that still uses the entire retrieval stack. This document specifies the method precisely and sets out the experiment that determines which of its ingredients is responsible.

2 Problem statement

2.1 Overton coverage

Let q be a contested question and U_q the population polled on it. A partition $\mathcal{C}_q = \{c_1, \ldots, c_{K_q}\}$ of U_q into *viewpoint clusters* groups participants who hold substantively the same position. Each participant u rates a candidate answer y on a five-point scale, $r(u, y) \in \{1, \ldots, 5\}$, according to how well y represents their own view.

Definition 1 (Representation)**.** A cluster c is *properly represented* by y when its members endorse the answer on average:

$$\mathrm{cov}(c, y) \;=\; \mathbf{1}\Bigg[\, \frac{1}{|c|} \sum_{u \in c} r(u, y) \;\geq\; \theta \,\Bigg], \qquad \theta = 4.$$

Definition 2 (Overton coverage)**.** For weights $w_c > 0$,

$$S_w(y, q) \;=\; \frac{\sum_{c \in \mathcal{C}_q} w_c \,\mathrm{cov}(c, y)}{\sum_{c \in \mathcal{C}_q} w_c},$$

and the task is to choose a policy $\pi : q \mapsto y$ maximising $\mathbb{E}_q\big[S_w(\pi(q), q)\big]$. The benchmark's own metric is $w_c \equiv 1$; we also report $w_c = \mathbf{1}[|c| < \frac{1}{2} \max_j |c_j|]$ and $w_c \propto |c|^{-1}$ as exploratory secondaries.

Remark 1 (Representation is self-certified)**.** The criterion is endorsement by the cluster's own members, not semantic similarity between y and a canonical statement of the cluster's position. An answer that *names* a position without articulating it can score high on similarity and still fail θ; we measure this directly, and a position-naming condition scored 0.072.

Remark 2 (One answer, not a set)**.** π emits a single string. The union over k independent samples, $\max_c$ taken sample-wise, upper-bounds what any single answer can achieve, and that bound is loose: unioning eight samples of one model exceeds a single sample by $+0.15$ coverage. The difficulty of the task is entirely in compressing that union into one answer, not in generating diverse content.

2.2 The coverage ledger

Maximising S_w against a reference answer y_0 is not the same as adding covered clusters, because coverage is not monotone in the content added. Define

$$G(y) = \{c : \mathrm{cov}(c, y) = 1,\ \mathrm{cov}(c, y_0) = 0\}, \qquad L(y) = \{c : \mathrm{cov}(c, y) = 0,\ \mathrm{cov}(c, y_0) = 1\},$$

the clusters gained and lost. Then exactly

$$S_w(y) - S_w(y_0) \;=\; \frac{\sum_{c \in G(y)} w_c \;-\; \sum_{c \in L(y)} w_c}{\sum_{c \in \mathcal{C}_q} w_c}.$$

Both terms must be reported. An intervention can move several times as much mass as another in both directions and tie it on net, and reporting G alone makes the two indistinguishable.

Under $w_c \equiv 1$ the prevalences of G and L are irrelevant: recovering a cluster held by 5% of the population exactly offsets dropping one held by 60%. This is the structural reason a method that trades majority representation for minority representation cannot profit under the benchmark's own metric, however well it performs the trade.

2.3 Why the trade is forced

The following is a model, not a measurement, and is offered because it accounts for the observed ledgers. Suppose cluster c requires d_c words of genuine articulation before its members endorse the answer, that y allocates $\ell_c(y)$ words to position c, and that the effective rating declines in the total length $L(y) = \sum_c \ell_c(y)$ through dilution. Then

$$\operatorname{cov}(c, y) \approx \mathbf{1}\big[\ell_c(y) \geq d_c\big] \cdot \mathbf{1}\big[L(y) \leq \Lambda\big],$$

and maximising S_w becomes a 0–1 knapsack: select a subset $T \subseteq \mathcal{C}_q$ maximising $\sum_{c \in T} w_c$ subject to $\sum_{c \in T} d_c \leq \Lambda$.

Two consequences match what we observe. First, an answer cannot add a position without either lengthening or taking words from another position, so G and L are coupled and gains are not free. Second, under uniform w the optimum is greedy on d_c, which favours positions that are cheap to articulate; there is no term rewarding the recovery of rare views, so any prevalence-aware method is optimising something the metric does not score. The empirical quantity standing in for Λ is judge-specific, which is why judge choice and length are treated as first-order concerns (Sections efsec:eval and efsec:confounds) rather than as robustness checks.

3 Method

3.1 The viewpoint graph

Let G be a graph whose leaves are *subgroup opinion nodes*. A leaf ℓ names a demographic group (e.g. an ideology or age band) and carries that group's empirical answer distribution $p_\ell \in \Delta^{m-1}$ over the m options of one survey item. Internal nodes are survey items; a leaf's parent is the item it answers.

Definition 3 (Fork)**.** A fork is a triple $f = (a, A, B)$: an anchor item a and two of its children A, B whose distributions differ. Forks are scored

$$s(f) \;=\; \max\big(\operatorname{rel}(q, a),\, 0\big)^{\alpha} \cdot W(p_A, p_B),$$

where $\operatorname{rel}(q, a)$ is the similarity of the anchor to the user's question q, gated at τ, and W is a Wasserstein distance whose ground cost comes from a hyperbolic embedding of G. Forks are returned in decreasing s.

3.2 The opinion axis

Given a fork $f = (a, A, B)$, let $L(a)$ be the children of a carrying distributions. If $|L(a)| < 3$ the anchor has no usable spectrum and the method degrades to a plain answer (Section 3.6). Otherwise define the pole-to-pole direction $v = p_B - p_A$ and the scalar coordinate

$$t(\ell) \;=\; \frac{\langle p_\ell - p_A,\; v \rangle}{\|v\|^2},$$

so that $t(A) = 0$ and $t(B) = 1$. Sorting $L(a)$ by t orders the subgroups along a genuine opinion axis rather than an arbitrary index. Note that a subgroup may satisfy $t < 0$ or $t > 1$: the extremes of the sorted list are not necessarily A and B.

3.3 Persona selection

Given a budget of k personas, take the two endpoints of the sorted axis, then repeatedly split the widest remaining gap. Endpoints first because they are furthest apart and therefore most likely to make the drafts disagree, which is the only thing the merge can harvest; the middle next because an audit of non-pole subgroups found 68% of them lie genuinely between the poles rather than being blends of them. At the default $k = 2$ the selection is exactly the two endpoints.

3.4 Draft generation

With n drafts total:

- **Draft 1** is a plain answer: the bare instruction *"Answer the question thoughtfully and concisely"* with no retrieved content. It is identical to the baseline arm, and is kept because the plain answer is the strongest single condition we have measured; trading it for one more persona trades the best draft for a marginal one.
- **Drafts** $2 \dots n$ each receive exactly one subgroup, rendered as its label and its real answer percentages (options below 5% suppressed), followed by the question. The instruction asks the model to argue the question as that group would, to commit to the view without hedging, to present no other group's position, and not to mention the group, the survey, or the instruction.

All drafts are sampled at temperature 0.7.

Remark 3. The anchor is a *related* survey item, not the benchmark question. A persona is therefore "the view of a group that answered a neighbouring question this way," not that group's answer to the question being asked. This is a source of topic drift and is one of the things the controls in Section 4 separate.

3.5 Merging

A single call assembles the drafts: one paragraph per distinct position, each carried at the length of the draft that explained it best, with disagreeing positions neither collapsed nor reconciled.

A guard rejects compression. Writing W_m, P_m for the merged answer's words and positions (paragraphs of at least 20 words) and W_d, P_d for the maxima over drafts, the merge is accepted iff $W_m \geq W_d$ and $P_m \geq P_d$; otherwise the drafts are concatenated verbatim. An earlier merge without this guard lost to the plain answer by summarising, so the fallback rate is reported as a measure of how often the model cannot merge losslessly.

3.6 Degradation

When the anchor has fewer than three subgroups with distributions, no personas are selected and the arm emits the plain draft alone. Such a row is a baseline answer wearing the condition's name, and dilutes the measured effect toward zero. The number of personas actually used is recorded per row so that the dilution can be quantified and, if needed, the effect re-estimated on treated rows only.

4 Experiment design

persona_merge holds five things at once: multi-sample generation, a merge, viewpoint-conditioned drafts, real survey distributions, and a relevance- and divergence-ranked anchor. The following arms remove one ingredient each, forming a ladder in which the gap between consecutive rungs isolates one factor.

Arm	Drafts	Viewpoints chosen by	Isolates
`baseline`	1	—	—
`plain_merge`	n plain, T=0.7	—	multi-draft assembly
`persona_self`	$1 + v$	the model	the graph's selection
`persona_rand`	$1 + k$	random anchor	the divergence ranking
`persona_merge`	$1 + k$	top-ranked anchor	(full method)

plain_merge. n independent samples of the plain answer at temperature 0.7, assembled by the same merge. No graph content of any kind. This is the control with the most at stake: the measured headroom from unioning 8 samples of one model against 1 is +0.15, larger than any arm difference we have recorded and stable across the judge swap. persona_merge is three samples plus a merge, so a large part of its effect could be that alone.

persona_self. The model is asked to name the genuinely distinct perspectives on the question—what each view is, roughly who holds it, and the consideration driving it—one per line, as many as it judges to exist. Each named perspective then conditions one draft, under an instruction matched to the graph version. This replaces the graph's subgroup selection with the model's own guess while holding the draft structure fixed.

Two asymmetries are deliberate and recorded rather than removed. The selection call is extra compute, and the arm is not pinned to the graph's two poles: it uses as many viewpoints as it names, up to a compute cap. Both are logged per row, so a win can be re-estimated at matched viewpoint count. The perspectives are stated at comparable richness to the graph's rendering (who holds the view, not only what it is), because a one-line slogan would hand this arm a thinner vantage point than the graph gives, biasing the comparison in the opposite direction to the compute advantage. Two biases of opposite sign would leave the result with no readable direction.

persona_rand. Identical to persona_merge except the anchor is drawn uniformly from the retrieved forks rather than taken as the top-scored one. Same relevance gate, same graph, same axis construction, same merge. This is the first arm in which the divergence ranking has a direct path to the output: under instruction-varied merging every draft sees every subgroup, so which fork ranked first barely propagates, whereas here the anchor determines which two subgroups the drafts argue from.

Anchors are not filtered to those with usable spectra, because persona_merge does not filter either; filtering would match the arms on fallback rate while breaking the comparison. Fallback rates are compared post hoc instead.

5 Evaluation

Benchmark. 60 contested questions, each with a population partitioned into viewpoint clusters. A judge predicts, for each participant, the 1–5 rating that participant would give the answer; a cluster counts as covered when its members' mean predicted rating is at least 4. An arm's score on a question is the fraction of that question's clusters it covers. To bound cost, 20 participants are sampled per question.

Judge. Judge choice is not a detail here: it reverses conclusions. Two judges were compared on the axes that matter for this metric. Against held-out human ratings, the first shows a length bias rising from +0.147 to +0.333 across length strata and marks $P(\text{rating} \geq 4) = 0.747$; the second shows −0.017 and 0.647, against human values of ≈ 0 and 0.650. We therefore report the second, and report the first only to show which conclusions are judge-dependent. Judges are run with reasoning disabled and ratings parsed strictly, because a leaked reasoning trace containing a digit would otherwise be read as a rating.

Primary metric. Uniform cluster coverage, the benchmark's own definition, is what we quote.

Secondary metrics, declared exploratory. Uniform weighting counts a view held by 5% of participants exactly as much as one held by 60%. Since the mechanism under test is a prevalence trade, we also report coverage restricted to clusters below half the size of the question's largest, and coverage weighted by inverse relative prevalence. These were adopted after seeing uniform results and are reported as exploratory; they are not used to support a primary claim.

Decomposition. Because the score is a difference of two quantities that move independently, we report, against a stated reference arm: clusters gained per question, clusters lost per question, and the mean relative prevalence of each set. An arm's net score is reproduced exactly by this decomposition. Reporting gains alone—as we initially did—conceals arms that tie on net while moving several times as much mass in both directions.

Inference. All comparisons are paired on question. For each arm we compute a per-question coverage difference against the reference, then bootstrap 10,000 resamples *over questions*, reporting the 2.5th and 97.5th percentiles. Questions, not clusters, are the resampling unit: clusters within a question share one answer and one difficulty, so treating them as independent would inflate the sample size roughly fivefold. The reported p is the inversion of that interval and carries no information the interval does not.

Stated limitations of the inference. No correction is applied for multiple comparisons, and each run yields several arm × weighting tests; we note which results would survive a family-wise correction. The intervals capture question-sampling variance only: judge error is not propagated, and comparisons *across* judges have no interval at all. Minority-weighted comparisons run on the subset of questions possessing a minority cluster, a slightly different question set from the uniform comparisons.

6 Controls on confounds

Length. The judge penalises length, and arm lengths differ by an order of magnitude, so length is reported per arm and any ordering is checked for monotonicity in it. In the run motivating this design, score was *not* monotone in length: the winning arm was longer than a losing one and 6.6× longer than the plain answer it beat. A length artifact cannot produce a non-monotone ordering, but per-question length regressions within arm remain the stronger test and are reported where available.

Compute. Arms differ in calls per answer. Counts are stated with every result, and any arm that wins while spending more is re-run at matched budget before the size of its advantage is interpreted.

Sampling. All arms within a comparison use the same number of rollouts, since the judge unions coverage over rollouts and an arm with fewer samples would be scored on a smaller union. Random-anchor arms in particular need more than one draw per question, or the selection rule is confounded with the luck of the draw.

One judging pass. Participant subsampling means a second judging run draws a different panel and shifts absolute scores. Arms are therefore judged together in a single pass, and scores from separate passes are never compared.

7 What each outcome licenses

- **plain_merge ≈ persona_merge.** The effect is multi-sample assembly. Neither the graph nor the personas contribute, and the remaining arms answer a question that no longer matters. This is the outcome that would most change the project's direction.
- **persona_self ≈ persona_merge.** The graph's contribution is structural, not informational: what matters is that drafts are assigned opposing viewpoints, and the model already knows what the disagreement is. Real subgroup distributions add nothing beyond prompting the model to take sides.
- **persona_merge > persona_self.** Empirical subgroup distributions carry viewpoint information the model does not generate unprompted. This would be the first direct evidence in this project that the graph contributes *content*.
- **persona_rand ≈ persona_merge.** Relevance and divergence ranking do not matter even where they have a direct path to the output, consistent with every selection ablation to date, and the scout reduces to a source of arbitrary opposed subgroups.
- **persona_merge > persona_rand.** The ranking does work, and previously null selection ablations were null because the instruction-varied draft scheme could not transmit the anchor choice—not because the ranking is uninformative.

A replication of persona_merge itself is obtained as a by-product, since it is regenerated alongside the controls; its current support is a single generation run scored by two judges.

8 Threats to validity

The benchmark separates systems but not samples of one system: a judge ranking eight samples of a single model against an independent second pass of itself achieves 0.507 and 0.328 concordance under the two judges, at or below chance. Any per-sample selection or reward signal is therefore unvalidatable on this benchmark, which bounds what these results can be extended to.

Coverage deltas are sensitive to judge strictness in a way that is not captured by any interval reported here: the share of clusters reached by no arm moves from 20% to 47% on judge choice alone. Absolute coverage should not be compared across judges, and we do not.

Finally, all cluster attributions are set arithmetic over judge verdicts. Nothing here verifies that a cluster covered by an injected arm was covered *because of* the injected content; a longer, more discursive answer may reach it incidentally. Establishing that link requires tracing retrieved content through drafts into the final answer, which is instrumented but not yet run.